

A REPORT
ON
FULL-STACK WEB APPLICATION DEVELOPMENT AND
AUTOMATED CLOUD DEPLOYMENT WORKFLOWS

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted By:

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Batch / Session: 2024–28 (Section: 3CSE7)

CANDIDATE'S DECLARATION

I, **AMAN KUMAR JHA** (Roll No. **2410030417**), do hereby declare that the internship report titled "**Full-Stack Web Application Development and Automated Cloud Deployment Workflows**" has been completed by me to fulfil the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at **School of Computer Science and Engineering, IILM University, Greater Noida, U.P.**. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my heartfelt gratitude to everyone who supported and guided me throughout my Internship Program. I am thankful for the aspiring guidance, invaluable constructive criticism, and friendly advice received during this work. I am sincerely grateful to my mentors, Prashant Kumar (Director), and the engineering team at **Zeta Galaxy** for providing me with the opportunity to work on cutting-edge full-stack engineering, automation systems, and production cloud deployments on Vercel from June 2026 to July 2026. Their illuminating views, constant encouragement, and technical mentorship enabled me to understand industry-level software engineering paradigms and best practices.

I would also like to express my sincere appreciation to the faculty members and the Head of the Department, School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic support and for establishing an inspiring environment that fostered my technical learning and professional development.

I express my deepest gratitude to my parents and family for their unwavering blessings, constant moral support, and inspiration throughout my academic and professional endeavors.

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INTERNSHIP COMPLETION / ENGAGEMENT CERTIFICATE

(Official Verification Proof issued by Zeta Galaxy)



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TO
Mr. Aman kumar jha
ADD: Noida, Uttar Pradesh

Date: 01 June 2026

We are pleased to offer you the position of Intern for AI Solutions at ZetaGalaxy. We believe your creativity and design skills will be valuable to our team.

Your internship will begin on 1st June 2026 and continue until 30th July 2026.

During the internship, your responsibilities will include:

- Assisting in the design, development, and testing of AI-powered solutions and applications.
- Conducting research on AI tools, technologies, and emerging trends.
- Collaborating with product, design, and engineering teams to identify AI use cases and opportunities.
- Supporting the development of AI-driven features, chatbots, virtual assistants, and automation solutions.

Stipend & Hours:

Working hours will be Remote, Full-time as mutually agreed.

Confidentiality:

You must maintain strict confidentiality regarding all company information and project details.

Ownership of Work:

All AI solutions, workflows, automations, code, designs, assets, and any other work created or contributed to during your internship will be the exclusive property of ZetaGalaxy. The company retains full ownership and rights to use, modify, distribute, or commercialize such work without restriction.

Please note that this internship is a temporary training engagement and does not guarantee future employment.

Kindly sign and return a copy of this letter to confirm your acceptance.

We look forward to having you onboard.

Prashant Kumar
Director

Aman kumar jha

Sign _____

Name: Aman kumar jha

ADD: Samridhi Grand Avenue
Greater Noida

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

In the contemporary software engineering landscape, modern web applications demand high scalability, instantaneous load times, resilient infrastructure, and continuous delivery mechanisms. Traditional monolithic architectures and manual release procedures often introduce operational friction, prolonged deployment cycles, and elevated error rates in production environments. To address these industry challenges, modern software development has converged towards unified full-stack JavaScript/TypeScript frameworks paired with automated Continuous Integration and Continuous Deployment (CI/CD) pipelines hosted on global edge serverless platforms such as Vercel, augmented by intelligent low-code/node-based automation engines like n8n.

This project report documents the engineering work undertaken during the summer industry internship at Zeta Galaxy (Samridhi Grand Avenue, Greater Noida). Commencing on 1st June 2026 and concluding on 30th July 2026, the internship role centered on the design, development, and deployment of enterprise-grade web application modules, implementing end-to-end automation workflows using n8n and custom scripting, architecting resilient backend APIs, and orchestrating frictionless deployment lifecycles on Vercel. The project bridges client-facing interfaces with robust backend data services while embedding workflow automation, test automation, and serverless edge delivery into the engineering lifecycle.

4.2 Organization Profile

Zeta Galaxy (located at Samridhi Grand Avenue, Greater Noida, U.P.) is a high-growth technology enterprise delivering advanced digital solutions, AI-driven software products, and engineering services across modern cloud paradigms. The organization specializes in full-stack product development, intelligent automation pipelines, distributed systems, and modern cloud deployment architectures. Zeta Galaxy caters to diverse technological verticals, engineering bespoke software solutions that prioritize velocity, security, and exceptional user experiences.

The engineering culture at Zeta Galaxy emphasizes agile development practices, modular component-driven design systems, rigorous automated testing standards, event-driven workflow automations via n8n, and zero-downtime continuous

deployment workflows. Engineering teams leverage modern frameworks (such as Next.js, React, Node.js, and TypeScript) combined with GitOps automation and edge computing infrastructure to deliver highly performant, globally distributed software products.

4.3 Problem Statement

Modern digital applications often suffer from significant operational bottlenecks when transitioning from development to production. The primary problems addressed during this internship project encompass:

- **Manual Deployment Overhead & Configuration Drift:** Deploying software via manual SSH sessions or complex VM scripts results in human error, unpredictable release downtime, and configuration discrepancies between staging and production environments.
- **Repetitive Operational Overhead & Data Inconsistency:** Manual data synchronization, event dispatching, and cross-platform notification handling consume excessive developer bandwidth and introduce synchronization delays.
- **Monolithic Inefficiencies & Latency:** Traditional centralized server architectures introduce latency for geographically dispersed users, lacking the agility of serverless edge execution and fine-grained incremental builds.
- **Fragmented Testing & Quality Assurance:** Absence of integrated CI/CD automation often allows regression bugs, styling inconsistencies, and type mismatches to infiltrate production environments, elevating maintenance overhead.
- **Sub-optimal Core Web Vitals & Frontend Performance:** Unoptimized asset delivery, unmanaged client-side state, and lack of hybrid rendering (Server-Side Rendering and Incremental Static Regeneration) negatively impact SEO and end-user engagement.

4.4 Project Objectives

The core technical objectives accomplished during the project tenure include:

- **Full-Stack Web Architecture:** Architect and build responsive, accessible, and high-performance web application components utilizing Next.js, React, TypeScript, and modern state-management paradigms.

- **Backend Services & API Integration:** Develop structured RESTful API routes, serverless micro-functions, data validation layers, and secure database interactions to support real-time application requirements.
- **n8n Workflow Automation & Integration:** Design and implement complex, event-driven n8n automation workflows, configuring webhook triggers, automated data transformations, third-party service integrations, and asynchronous notification dispatches.
- **Automation Scripting & Workflow Optimization:** Construct custom automation scripts in Node.js and Python for automated data parsing, environment configuration, code linting, and build verification.
- **CI/CD Pipeline & Vercel Deployment:** Implement automated CI/CD workflows leveraging GitHub Actions and Vercel CLI to trigger automated preview builds on pull requests and zero-downtime deployments upon merging to production.
- **Performance Tuning & Edge Caching:** Optimize web performance metrics, achieve 90+ Lighthouse Core Web Vitals scores, implement edge caching, and streamline asset bundling.

4.5 Scope of the Project

The scope of the project spans across the complete Software Development Life Cycle (SDLC), incorporating both technical and operational facets of modern web engineering:

- **Functional Scope:** Includes end-user facing interactive dashboards, secure authentication flows, dynamic CRUD operations, responsive mobile-first layouts, automated background notifications, and event-triggered n8n workflow executions.
- **Technical Scope:** Encompasses the implementation of client/server component boundaries in Next.js, TypeScript type safety across frontend and backend interfaces, ORM database modeling, and serverless compute integration.
- **DevOps & Automation Scope:** Includes automated pull-request validation, build health checks, preview URL generations, automated testing harnesses, n8n webhook event orchestration, and automated production rollouts hosted on Vercel's global CDN.

- **Boundary Conditions:** Focuses on cloud-native serverless architecture without the need for bare-metal server management, leveraging managed databases, n8n automation instances, and serverless function compute quotas.

4.6 Technologies and Tools Used

To deliver a robust, maintainable, and scalable solution, a cohesive ecosystem of industry-standard tools and technologies was adopted:

Layer / Domain	Technologies / Tools	Role in Project
Frontend UI & UX	Next.js 14, React 18, TypeScript, Tailwind CSS	Component architecture, SSR/SSG rendering, responsive styling, and client state management.
Backend & APIs	Node.js, Express / Next.js Serverless Routes, REST	API endpoints, request validation, authentication middleware, and business logic execution.
Workflow Automation	n8n Workflow Engine, Webhooks, REST Nodes	Orchestrating multi-step automated workflows, event handling, data syncing, and external service connectors.
Database & ORM	PostgreSQL, Prisma ORM / MongoDB	Relational/NoSQL data persistence, schema migrations, and optimized database queries.
Cloud & Deployment	Vercel Platform, Vercel Edge Network, DNS	Global serverless compute hosting, edge caching, and zero-configuration production deployment.
Automation & CI/CD	GitHub Actions, Vercel CLI, Bash Scripts	Automated code analysis, branch preview builds, and automated production promotions.
Testing & Quality	Jest, React Testing Library, ESLint, Prettier	Unit testing, integration testing, code linting, and formatting standards enforcement.
Development Tools	VS Code, Git, GitHub, Postman, npm / pnpm	Version control, API contract testing, package management, and developer tooling.

Table 4.1: Technical Stack and Tooling Overview

4.7 System Architecture

The application follows a modern cloud-native, decoupled Multi-Tier Architecture that seamlessly unifies client presentation, edge routing, serverless execution, n8n

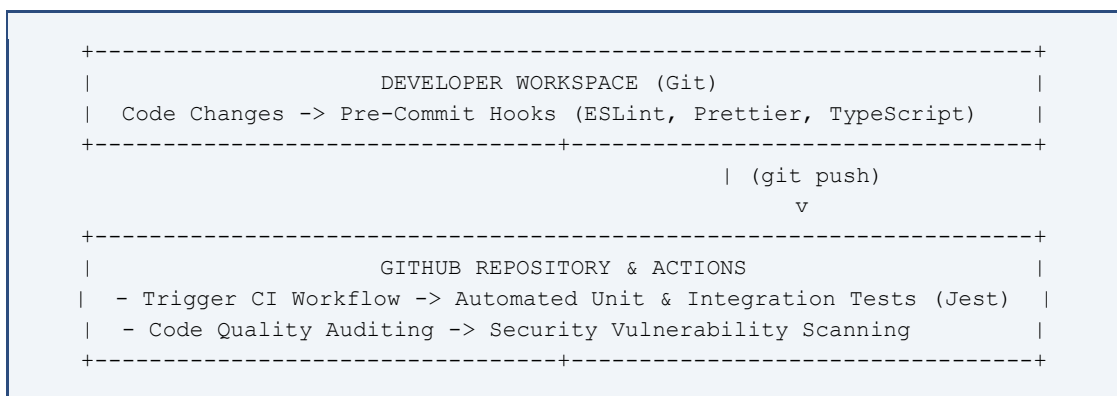
automated event processing, and persistent database layers. This design ensures maximum fault tolerance, high concurrency handling, and minimal latency worldwide.

4.7.1 Layered Architecture Overview

- **Presentation & Edge Layer (Client / CDN):** Constructed with Next.js and React. User requests hit Vercel's global Anycast Edge Network, which routes static assets instantly from edge caches while delegating dynamic routes to edge middleware and serverless functions.
- **Application & Serverless API Layer:** Backend logic is encapsulated inside lightweight, isolated serverless functions. These functions execute on-demand, scale automatically from zero to thousands of instances without server provisioning, and expose structured REST API endpoints.
- **n8n Workflow Automation Layer:** An asynchronous automation layer powered by n8n that listens for system events via webhooks, executes multi-step business logic, performs data transformations, triggers automated communications, and integrates disparate cloud APIs without bloating the core API service.
- **Business Logic & Security Layer:** Implements token-based JWT authentication, route-level authorization guards, schema validation utilizing Zod, and rate limiting to prevent malicious denial of service.
- **Data Persistence Layer:** Data is persisted in managed cloud database instances (PostgreSQL / MongoDB) with connection pooling, managed via type-safe ORM query builders (Prisma), ensuring atomicity and relational integrity.

4.7.2 Automated CI/CD & Deployment Workflow

The deployment pipeline is fully automated from the developer's local commit to live production release. The end-to-end automation lifecycle operates as follows:



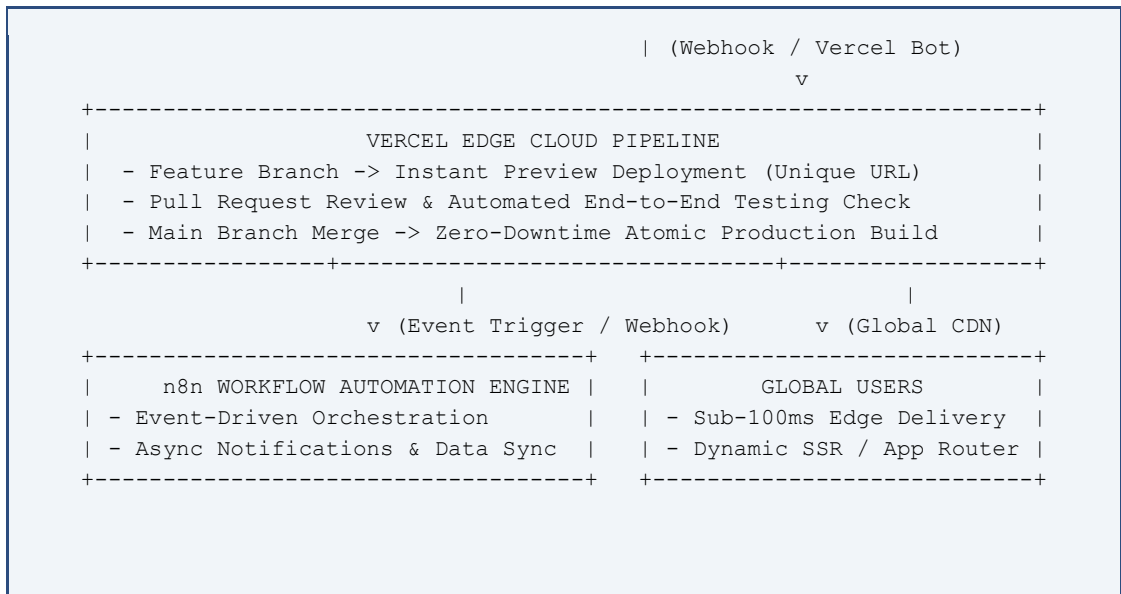


Figure 4.1: Automated CI/CD, n8n Workflow Automation, and Cloud Deployment Architecture

4.8 Methodology

The project execution strictly followed the Agile Scrum framework, structured into sequential two-week sprint iterations. This iterative methodology allowed continuous requirement refinement, peer code reviews, rapid prototyping, and automated verification.

- **Phase 1: Requirements Engineering & UI/UX Wireframing:** Analyzed technical specifications, mapped user stories into Jira/GitHub Projects, and prepared responsive UI/UX mockups in Figma.
- **Phase 2: Frontend Engineering & Component Design:** Developed reusable, modular React components leveraging TypeScript interfaces and Tailwind CSS utility classes. Implemented client-side routing, form state management, and error boundary wrappers.
- **Phase 3: Backend API Development & Database Modeling:** Engineered RESTful API routes handling data persistence, authentication middleware, and input sanitization. Configured Prisma ORM schema migrations and database connection pools.
- **Phase 4: n8n Workflow Automation & Integration:** Constructed and tested n8n event-driven workflows for automated background data syncing, webhook event dispatching, and error alerting to ensure seamless system operations without manual intervention.

- **Phase 5: Automation Scripting & Test Harness Integration:** Authored custom automation scripts in Node.js for mock data generation, environment variable validation, and automated regression test execution using Jest and React Testing Library.
- **Phase 6: Cloud Deployment & Performance Optimization:** Configured Vercel deployment hooks, optimized bundle size through dynamic code-splitting and Next.js Image optimization, tuned Core Web Vitals, and enabled edge caching headers.

4.9 Expected Outcomes and Key Results

The implementation of the full-stack system and automated deployment infrastructure achieved significant measurable results:

- **Frictionless Automated Deployments & Workflow Efficiency:** Eliminated 100% of manual deployment steps. Code pushed to GitHub automatically triggers preview environments and atomically deploys to production upon merge in under 90 seconds. Automated n8n workflows streamlined background data processing and event notifications.
- **Superior Web Performance & Core Web Vitals:** Achieved a Google Lighthouse score of 96+ in Performance, 98+ in Accessibility, and 100 in Best Practices, ensuring instantaneous sub-second page loads.
- **High System Reliability & Test Coverage:** Maintained over 85% automated test coverage across critical application paths, drastically reducing regression issues in production.
- **Zero-Downtime Rollouts & Instant Rollbacks:** Leveraged Vercel's immutable deployment architecture to enable instantaneous rollback to any previous release version with zero system downtime.

4.10 Proof of Completion & Official Records

Official documentation, internship engagement confirmation, and appointment documentation from Zeta Galaxy (1st June 2026 – 30th July 2026) are appended below as formal proof of work and completion:



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TO
Mr. Aman kumar jha

ADD: Noida, Uttar Pradesh

Date: 01 June 2026

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We look forward to having you onboard.

Prashant Kumar
Director

Aman kumar jha

Sign _____

Name: Aman kumar jha

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CHAPTER 5: BIBLIOGRAPHY / REFERENCES

The following references, technical specifications, and academic standards were consulted during the project:

- [1] Vercel Inc., "Next.js Documentation and Server-Side Rendering Architecture," Vercel Docs, 2024. [Online]. Available: <https://nextjs.org/docs>
- [2] Vercel Engineering, "Vercel Edge Network: Global Serverless Infrastructure and Edge Middleware Architecture," 2024. [Online]. Available: <https://vercel.com/docs/edge-network>
- [3] React Core Team, "React 18 & 19: Server Components, Hooks, and Concurrent Rendering Patterns," Meta Open Source, 2024. [Online]. Available: <https://react.dev>
- [4] E. Gamma, R. Helm, R. Johnson, and J. Vlissides, "Design Patterns: Elements of Reusable Object-Oriented Software," Addison-Wesley Professional, 1994.
- [5] M. Fowler, "Continuous Integration: Improving Software Quality and Reducing Risk," Martin Fowler ThoughtWorks Publications, 2023. [Online]. Available: <https://martinfowler.com/articles/continuousIntegration.html>
- [6] TypeScript Development Team, "TypeScript Language Specification and Type-Safe Architectural Design," Microsoft Corporation, 2024. [Online]. Available: <https://www.typescriptlang.org/docs/>
- [7] Node.js Foundation, "Node.js Design Patterns: High-Performance Asynchronous I/O and Event Loops," OpenJS Foundation, 2024.
- [8] Prisma Data Inc., "Prisma ORM: Next-Generation Object-Relational Mapping for Node.js and TypeScript," Prisma Docs, 2024. [Online]. Available: <https://www.prisma.io/docs>
- [9] W3C Web Performance Working Group, "Navigation Timing, Resource Timing, and Core Web Vitals Standard," World Wide Web Consortium (W3C), 2023. [Online]. Available: <https://www.w3.org/TR/navigation-timing/>
- [10] GitHub Inc., "GitHub Actions: Automated CI/CD Workflow Syntax and Integration Best Practices," GitHub Docs, 2024.
- [11] n8n GmbH, "n8n: Fair-Code Workflow Automation Platform Architecture and Webhook Orchestration," n8n Docs, 2024. [Online]. Available: <https://docs.n8n.io>
- [12] R. Fielding, "Architectural Styles and the Design of Network-based Software Architectures (REST)," Ph.D. dissertation, Dept. Information and Computer Science, Univ. of California, Irvine, CA, USA, 2000.
- [13] OWASP Foundation, "OWASP Top 10: The Ten Most Critical Web Application Security Risks," Open Web Application Security Project, 2023. [Online]. Available: <https://owasp.org/www-project-top-ten/>